

PY1010 — Physics for Biological and Chemical Sciences

Summer 2023 Examination · Paper 1 · Full Marking Scheme

Module	PY1010 — Physics for Biological and Chemical Sciences
Internal examiners	Prof. P. Callanan, Prof. A. Ruth
Institution	University College Cork
Session	Summer 2023, Paper 1
Duration	3 hours (+ 15 minutes reading time)
Structure	Six questions, each given as an either/or choice (A or B). Answer one option of each of the six questions and all parts of that option. Marks for each part are shown in square brackets.
Materials	Non-programmable calculator permitted; constants on the paper's "useful data" page

Coverage note. This scheme works through **both options (A and B) of all six questions** in full. On the day you only choose one option per question, but working both lets you pick the one you're strongest on.

Source basis. Every answer is built **only** from the PY1010 lecture material given to students: the lecture slides (mechanics, fluids, thermodynamics) and the handwritten lecture notes (gas laws, electricity, magnetism, waves and optics). No textbook or outside material is used, except where a highlighted note says the lecture notes don't cover something (Q5(A)(a) and Q6(B)).

Own-figure rule. Where a part carries forward a number from an earlier part, an answer that uses the correct method on an earlier wrong value still earns the method marks for the later part.

Numerical conventions. $g = 9.81 \text{ m s}^{-2}$ and other constants from the paper's useful-data page. Intermediate values are kept unrounded, and final answers are given to 2–3 significant figures. Answers differing only through rounding should receive full marks.

Study-Uni

Question 1 · Option A

Volumes, areas and mass of a year's rainfall over Ireland — 10 marks

Given: Annual rainfall 1151 mm, assumed over all of Ireland; area of Ireland 84 421 km²; raindrop radius 1 mm; $\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$.

1(A)(a)

[3 marks]

What volume of rain fell over every square kilometre last year? Express your answer in m³.

ANSWER

$1.151 \times 10^6 \text{ m}^3 \text{ per km}^2$

WORKING

The rain forms a layer 1151 mm deep over each km², like a very flat cylinder: volume = area × height.

Convert both to metres: 1151 mm = 1.151 m, and 1 km² = (10³ m)² = 10⁶ m².

$$V = 10^6 \text{ m}^2 \times 1.151 \text{ m} = 1.151 \times 10^6 \text{ m}^3$$

Marks	Awarded for
1	Depth in metres (1.151 m).
1	1 km ² = 10 ⁶ m ² .
1	V = 1.151 × 10 ⁶ m ³ .

1(A)(b)**[2 marks]**

If the total area of Ireland is 84 421 km², what total volume of rain fell on the country last year?

ANSWER

$$V_{\text{total}} \approx 9.7 \times 10^{10} \text{ m}^3$$

WORKING

$$V_{\text{total}} = 84\,421 \text{ km}^2 \times 1.151 \times 10^6 \text{ m}^3 \text{ per km}^2 = 9.72 \times 10^{10} \text{ m}^3$$

Marks	Awarded for
1	Multiplies by the area in km ² .
1	$\approx 9.7 \times 10^{10} \text{ m}^3$ (own-figure from (a) accepted).

1(A)(c)**[3 marks]**

If each raindrop can be approximated as a sphere of radius 1 mm, how many raindrops fell on Ireland last year?

ANSWER

$N \approx 2.3 \times 10^{19}$ raindrops

WORKING

Volume of one drop (sphere), with $r = 1 \text{ mm} = 10^{-3} \text{ m}$:

$$V_{\text{drop}} = (4/3) \pi r^3 = (4/3) \pi (10^{-3})^3 = 4.19 \times 10^{-9} \text{ m}^3$$

$$N = V_{\text{total}} / V_{\text{drop}} = 9.72 \times 10^{10} \div 4.19 \times 10^{-9} = 2.3 \times 10^{19}$$

Marks	Awarded for
1	Sphere volume with r in metres.
1	$N = V_{\text{total}}/V_{\text{drop}}$.
1	$\approx 2.3 \times 10^{19}$.

1(A)(d)**[2 marks]**

The density of water is 1000 kg m^{-3} . What was the total mass of rain water falling on Ireland last year?

ANSWER

$$m \approx 9.7 \times 10^{13} \text{ kg}$$

WORKING

$$m = \rho V = 1000 \text{ kg m}^{-3} \times 9.72 \times 10^{10} \text{ m}^3 = 9.7 \times 10^{13} \text{ kg}$$

Marks	Awarded for
1	$m = \rho V$.
1	$\approx 9.7 \times 10^{13} \text{ kg}$.

Question 1(A) Total: 10 Marks

Question 1 · Option B

Constant acceleration: Usain Bolt's 200 m — 10 marks

Given: 200 m in 19.19 s.

1(B)(a)

[3 marks]

If Usain Bolt crossed the finish line after a period of constant acceleration from the start, what would this acceleration have been, in m s^{-2} ?

ANSWER

$$a \approx 1.09 \text{ m s}^{-2}$$

WORKING

He starts from rest ($u = 0$) and covers $s = 200 \text{ m}$ in $t = 19.19 \text{ s}$. Use $s = ut + \frac{1}{2}at^2$:

$$200 = 0 + \frac{1}{2} a (19.19)^2 \Rightarrow a = 2 \times 200 \div 19.19^2$$

$$a = 400 \div 368.3 = 1.09 \text{ m s}^{-2}$$

Marks	Awarded for
1	$u = 0$ and $s = ut + \frac{1}{2}at^2$ chosen.
1	Rearranged for a .
1	$a \approx 1.09 \text{ m s}^{-2}$.

1(B)(b)**[3 marks]**

Using the acceleration from (a), what would his speed have been as he crossed the finish line, in m s^{-1} ?

ANSWER

$$v \approx 20.8 \text{ m s}^{-1}$$

WORKING

$$v = u + a t = 0 + 1.086 \times 19.19 = 20.8 \text{ m s}^{-1}$$

Check: $v^2 = 2as = 2 \times 1.086 \times 200 = 434.5$, so $v = 20.8 \text{ m s}^{-1}$.

Marks	Awarded for
1	Suitable equation ($v = u + at$ or $v^2 = u^2 + 2as$).
1	Correct substitution.
1	$v \approx 20.8 \text{ m s}^{-1}$ (own-figure accepted).

1(B)(c)**[4 marks]**

He starts the last 200 m leg of a relay at 5 m s^{-1} and accelerates at 1 m s^{-2} . What is his final speed at the finish line, and how long does the 200 m take?

ANSWER

$$v \approx 20.6 \text{ m s}^{-1}; t \approx 15.6 \text{ s}$$

WORKING

Final speed: $u = 5 \text{ m s}^{-1}$, $a = 1 \text{ m s}^{-2}$, $s = 200 \text{ m}$. Use $v^2 = u^2 + 2as$:

$$v^2 = 5^2 + 2 \times 1 \times 200 = 425 \Rightarrow v = 20.6 \text{ m s}^{-1}$$

Time: use $v = u + at$:

$$t = (v - u) / a = (20.6 - 5) \div 1 = 15.6 \text{ s}$$

Marks	Awarded for
1	$v^2 = u^2 + 2as$.
1	$v \approx 20.6 \text{ m s}^{-1}$.
1	$v = u + at$ rearranged for t .
1	$t \approx 15.6 \text{ s}$.

Question 1(B) Total: 10 Marks

Question 2 · Option A

Newton's law of gravitation, mass and weight, a diving kingfisher — 10 marks

2(A)(a)

[2 marks]

State and explain Newton's law of Gravitation. Give the SI units of the quantities in the law.

ANSWER

$F = Gm_1m_2/r^2$: attractive, $\propto$ product of masses, $\propto 1/\text{distance}^2$

WORKING

$$F = G m_1 m_2 / r^2$$

Every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

Units: F in N; m_1, m_2 in kg; r in m; $G = 6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$, the universal gravitational constant.

Marks	Awarded for
1	Law stated and explained.
1	SI units of F , m , r and G .

2(A)(b)(i)**[1 mark]**

How are the mass and the weight of an object related? Explain.

ANSWER

$$w = mg$$

WORKING

Weight is the gravitational force on a mass: $w = mg$. Mass (kg) is the amount of matter and is fixed. Weight (N) depends on the local g .

Marks**Awarded for****1** $w = mg$, with explanation.**2(A)(b)(ii)****[1 mark]**

You step from the ground onto a chair. Relative to the ground, does your weight increase, decrease, or remain the same while you stand on the chair?

ANSWER

It decreases, but only by a negligible amount (r increases slightly, so $g = GM_E/r^2$ falls)

WORKING

$g = G M_E/r^2$, where r is your distance from the centre of the Earth. On the chair r is slightly larger, so g , and therefore your weight, is very slightly smaller. The change is tiny: about half a metre compared with $r \approx 6378$ km.

Marks**Awarded for****1**Decreases (negligibly), explained via $g \propto 1/r^2$.

2(A)(c)(i)**[3 marks]**

A kingfisher on a branch 4 m above a pond dives using only gravity (negligible air resistance). How long does it take to reach the prey at the water surface?

ANSWER

$$t \approx 0.90 \text{ s}$$

WORKING

Free fall from rest ($u = 0$) with $a = g$, through $h = 4$ m. Use $x = \frac{1}{2}gt^2$:

$$t = \sqrt{(2h / g)} = \sqrt{(2 \times 4 \div 9.81)} = \sqrt{0.8155} = 0.90 \text{ s}$$

Marks	Awarded for
1	Free fall with $u = 0$, $a = g$.
1	$t = \sqrt{(2h/g)}$.
1	$t \approx 0.90 \text{ s}$.

2(A)(c)(ii)**[3 marks]**

How fast is the kingfisher when it hits the prey?

ANSWER

$v \approx 8.9 \text{ m s}^{-1}$

WORKING

$$v = \sqrt{2 g h} = \sqrt{2 \times 9.81 \times 4} = 8.9 \text{ m s}^{-1}$$

Check: $v = gt = 9.81 \times 0.903 = 8.9 \text{ m s}^{-1}$.

Marks	Awarded for
1	$v = \sqrt{2gh}$ or $v = gt$.
1	Correct substitution.
1	$v \approx 8.9 \text{ m s}^{-1}$.

Question 2(A) Total: 10 Marks

Question 2 · Option B

Archimedes' principle: a cube submerged in a fluid — 10 marks

Given: Cube of side 20.00 cm fully immersed; pressure 105 kPa at its top face, 107 kPa at its bottom face; air pressure 99 800 Pa.

2(B)(a)

[2 marks]

State Archimedes' Principle.

ANSWER

Buoyant force = weight of the fluid displaced

WORKING

When a body is completely or partially immersed in a fluid, the fluid exerts an **upward force** on the body equal to the **weight of the fluid displaced** by the body.

Marks	Awarded for
1	Upward force on an immersed body.
1	Equal to the weight of fluid displaced.

2(B)(b)(i)**[2 marks]**

What is the net upwards (buoyant) force acting on the cube?

ANSWER

$$F_{\text{buoy}} = 80 \text{ N (upwards)}$$

WORKING

The pressures on the four side faces cancel. The net force comes from the pressure difference between the bottom and top faces, each of area $A = (0.20 \text{ m})^2 = 0.04 \text{ m}^2$.

$$F = (P_{\text{bottom}} - P_{\text{top}}) \times A = (107\,000 - 105\,000) \text{ Pa} \times 0.04 \text{ m}^2 = 80 \text{ N}$$

Marks	Awarded for
1	Uses the pressure difference $\times$ face area.
1	80 N.

2(B)(b)(ii)**[2 marks]**

Use Archimedes' Principle to calculate the mass of the displaced fluid.

ANSWER

$$m_{\text{fl}} \approx 8.2 \text{ kg}$$

WORKING

The buoyant force equals the weight of the displaced fluid, $F = m_{\text{fl}} g$:

$$m_{\text{fl}} = F / g = 80 \text{ N} \div 9.81 \text{ m s}^{-2} = 8.15 \text{ kg}$$

Marks	Awarded for
1	$F = m_{\text{fl}} g$.
1	$\approx 8.2 \text{ kg}$.

2(B)(b)(iii)**[2 marks]**

What is the density of the displaced fluid?

ANSWER

$$\rho \approx 1020 \text{ kg m}^{-3}$$

WORKING

The displaced volume equals the cube's volume: $V = (0.20 \text{ m})^3 = 8.0 \times 10^{-3} \text{ m}^3$.

$$\rho = m / V = 8.155 \text{ kg} \div 8.0 \times 10^{-3} \text{ m}^3 = 1.02 \times 10^3 \text{ kg m}^{-3}$$

Marks	Awarded for
1	$V = 8 \times 10^{-3} \text{ m}^3$.
1	$\rho \approx 1020 \text{ kg m}^{-3}$.

2(B)(b)(iv)**[2 marks]**

What is the depth d of the cube below the surface of the fluid? Air pressure was 99 800 Pa.

ANSWER

$$d \approx 0.52 \text{ m}$$

WORKING

The pressure at depth d is the air pressure plus the hydrostatic pressure: $P_{\text{top}} = P_{\text{atm}} + \rho g d$.

$$d = (P_{\text{top}} - P_{\text{atm}}) / (\rho g) = (105\,000 - 99\,800) \div (1019 \times 9.81)$$

$$d = 5200 \div 10\,000 = 0.52 \text{ m}$$

Marks	Awarded for
1	$P = P_{\text{atm}} + \rho g d$.
1	$d \approx 0.52 \text{ m}$.

Question 2(B) Total: 10 Marks

Question 3 · Option A

Specific heat, latent heat of fusion, defrosting a freezer — 10 marks

Given: Hairdryer 1 kW; 1 kg of ice at $-10\text{ }^{\circ}\text{C}$; $c_{\text{ice}} = 2.22 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$; $L_f = 3.35 \times 10^5 \text{ J kg}^{-1}$.

3(A)(a)

[2 marks]

Define (i) specific heat capacity and (ii) heat of fusion.

ANSWER

$c = Q/(m\Delta T) \text{ [J kg}^{-1} \text{ K}^{-1}]$; $L_f = Q/m \text{ for melting at constant } T \text{ [J kg}^{-1}]$

WORKING

(i) **Specific heat capacity** c : the heat required per unit mass to change the temperature of a substance by 1 K. From $Q = mc\Delta T$; unit $\text{J kg}^{-1} \text{ K}^{-1}$.

(ii) **Heat of fusion** L_f : the heat required per unit mass to change a substance from solid to liquid at constant temperature (the melting point). From $L = Q/m$; unit J kg^{-1} .

Marks	Awarded for
1	Specific heat capacity defined.
1	Heat of fusion defined (solid $\rightarrow$ liquid at constant T).

3(A)(b)(i)**[2 marks]**

Calculate the amount of heat needed to raise the temperature of the ice to 0 °C.

ANSWER

$$Q_1 = 2.22 \times 10^4 \text{ J (22.2 kJ)}$$

WORKING

$$Q_1 = m c_{\text{ice}} \Delta T = 1 \text{ kg} \times 2.22 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1} \times 10 \text{ K} = 2.22 \times 10^4 \text{ J}$$

Marks	Awarded for
1	$Q = mc\Delta T$ with $\Delta T = 10 \text{ K}$.
1	22.2 kJ.

3(A)(b)(ii)**[2 marks]**

Calculate the amount of heat required to convert the ice at 0 °C to water at the same temperature.

ANSWER

$$Q_2 = 3.35 \times 10^5 \text{ J (335 kJ)}$$

WORKING

$$Q_2 = m L_f = 1 \text{ kg} \times 3.35 \times 10^5 \text{ J kg}^{-1} = 3.35 \times 10^5 \text{ J}$$

Marks	Awarded for
1	$Q = mL_f$.
1	335 kJ.

3(A)(b)(iii)**[4 marks]**

If 50% of the heat from the hairdryer heats the ice, how long will it take to melt all the ice?

ANSWER

$t \approx 714 \text{ s}$ (about 12 minutes)

WORKING

Total heat needed:

$$Q = Q_1 + Q_2 = 22\,200 + 335\,000 = 3.572 \times 10^5 \text{ J}$$

Useful power delivered to the ice: $50\% \times 1000 \text{ W} = 500 \text{ W}$. Since $P = Q/t$:

$$t = Q / P = 3.572 \times 10^5 \text{ J} \div 500 \text{ W} = 714 \text{ s} \approx 11.9 \text{ minutes}$$

Marks	Awarded for
1	Total heat $Q_1 + Q_2$.
1	Useful power = 500 W.
1	$t = Q/P$.
1	$\approx 714 \text{ s}$.

Question 3(A) Total: 10 Marks

Question 3 · Option B

Ideal gas law $PV = NkT$, Boyle's and Charles' laws, the spy balloon — 10 marks

Given: Balloon radius 10 m at 20 km altitude: $T = -58\text{ }^{\circ}\text{C}$, $P = 5000\text{ Pa}$. At 5 km altitude: $T = -13\text{ }^{\circ}\text{C}$, $P = 50\,000\text{ Pa}$.

3(B)(a)(i)

[2 marks]

Write down the SI units for P and T in $PV = NkT$.

ANSWER

P in Pa (N m^{-2}); T in K

WORKING

P is pressure: force per area, unit the pascal, $\text{Pa} = \text{N m}^{-2}$.

T must be the **absolute** temperature, in kelvin (K).

Marks	Awarded for
1	Pa.
1	K.

3(B)(a)(ii)

[1 mark]

Under what conditions will this expression lead to Boyle's Law?

ANSWER

Fixed amount of gas (N constant) at constant $T \Rightarrow PV = \text{constant}$

WORKING

For a fixed amount of gas (N constant) at constant temperature T , NkT is constant, so $PV = \text{constant}$. That is Boyle's law: $P \propto 1/V$.

Marks	Awarded for
1	N and T constant.

3(B)(a)(iii)**[1 mark]**

Under what conditions will this expression lead to Charles' Law?

ANSWER**Fixed amount of gas at constant $P \Rightarrow V/T = \text{constant}$** **WORKING**

For a fixed amount of gas at constant pressure P , $V = (Nk/P) T$, so $V/T = \text{constant}$. That is Charles' law, with T in K.

Marks	Awarded for
1	N and P constant.

3(B)(b)(i)**[3 marks]**

What is the number of gas molecules in the balloon?

ANSWER **$N \approx 7.0 \times 10^{27}$** **WORKING**

Volume of the sphere: $V = (4/3)\pi r^3 = (4/3)\pi(10 \text{ m})^3 = 4189 \text{ m}^3$. Temperature: $T = -58 + 273.15 = 215.15 \text{ K}$.

$$N = P V / (k T) = (5000 \times 4189) \div (1.381 \times 10^{-23} \times 215.15)$$

$$N = 7.0 \times 10^{27} \text{ molecules}$$

Marks	Awarded for
1	Sphere volume and T in K.
1	$N = PV/(kT)$.
1	$\approx 7.0 \times 10^{27}$.

3(B)(b)(ii)**[3 marks]**

Calculate the radius of the balloon at 5 km. Assume the inside of the balloon is always at the same temperature and pressure as the surrounding air.

ANSWER

$$r \approx 4.9 \text{ m}$$

WORKING

The amount of gas is fixed, so $P_1 V_1 / T_1 = P_2 V_2 / T_2$, with $T_2 = -13 + 273.15 = 260.15 \text{ K}$:

$$V_2 = V_1 (P_1 / P_2) (T_2 / T_1) = 4189 \times (5000 / 50\,000) \times (260.15 / 215.15) = 506 \text{ m}^3$$

$$r_2 = (3V_2 / 4\pi)^{1/3} = (120.9)^{1/3} = 4.9 \text{ m}$$

Marks	Awarded for
1	$P_1 V_1 / T_1 = P_2 V_2 / T_2$ with T in K.
1	$V_2 \approx 506 \text{ m}^3$.
1	$r \approx 4.9 \text{ m}$.

Question 3(B) Total: 10 Marks

Question 4 · Option A $v = f\lambda$, decibels, intensity and distance: hearing a pin drop — 10 marks

Given: A pin releases 10^{-11} J of sound energy over 1 ms, radiated equally in all directions.

4(A)(a)**[1 mark]**

Write down the formula that relates the frequency, wavelength and speed of a wave.

ANSWER

$$v = f\lambda$$

WORKING

$$v = f \lambda$$

v in m s^{-1} , f in Hz, λ in m.

Marks**Awarded for****1** $v = f\lambda$.

4(A)(b)**[1 mark]**

Write down the formula that relates a sound intensity in W m^{-2} to the sound intensity level in dB.

ANSWER

$$\text{dB} = 10 \log_{10}(I/I_0), \text{ with } I_0 = 10^{-12} \text{ W m}^{-2}$$

WORKING

$$\text{dB} = 10 \log_{10}(I / I_0)$$

$I_0 = 10^{-12} \text{ W m}^{-2}$ is the intensity at the threshold of human hearing.

Marks**Awarded for****1**Formula with I_0 identified.**4(A)(c)(i)****[2 marks]**

What is the rate, L , at which this energy is radiated, in W?

ANSWER

$$L = 1 \times 10^{-8} \text{ W}$$

WORKING

Power = energy ÷ time:

$$L = 10^{-11} \text{ J} \div 10^{-3} \text{ s} = 10^{-8} \text{ W}$$

Marks**Awarded for****1** $L = E/t$ with $1 \text{ ms} = 10^{-3} \text{ s}$.**1** 10^{-8} W .

4(A)(c)(ii)**[3 marks]**

Your ear can just hear $I = 10^{-12} \text{ W m}^{-2}$. At what maximum distance d from the pin could you hear it hit the floor? Use $L/(4\pi d^2) = I$.

ANSWER

$$d \approx 28 \text{ m}$$

WORKING

Rearrange for d :

$$d = \sqrt{L / (4\pi I)} = \sqrt{10^{-8} \div (4\pi \times 10^{-12})}$$

$$d = \sqrt{795.8} = 28 \text{ m}$$

Marks	Awarded for
1	Rearranged for d .
1	Correct substitution.
1	$d \approx 28 \text{ m}$.

4(A)(c)(iii)**[3 marks]**

If the frequency is 10 kHz and the wavelength 3 cm, how long does the sound take to travel from the pin to your ear? Use your result from (ii).

ANSWER

$t \approx 0.094 \text{ s (94 ms)}$

WORKING

$$v = f \lambda = 10^4 \text{ Hz} \times 0.03 \text{ m} = 300 \text{ m s}^{-1}$$

$$t = d / v = 28.2 \text{ m} \div 300 \text{ m s}^{-1} = 0.094 \text{ s}$$

Marks	Awarded for
1	$v = f\lambda = 300 \text{ m s}^{-1}$, with units converted.
1	$t = d/v$.
1	$\approx 0.094 \text{ s}$ (own-figure from (ii) accepted).

Question 4(A) Total: 10 Marks

Question 4 · Option B

Coulomb's law, the electric field, a charged water drop — 10 marks

4(B)(a)**[1 mark]**

State Coulomb's law for the interaction between two electric charges.

ANSWER

$$F = Qq/(4\pi\epsilon_0 r^2)$$

WORKING

$$F = Q q / (4\pi \epsilon_0 r^2)$$

The force between two point charges Q and q is proportional to the product of the charges and inversely proportional to the square of their separation r . Like charges repel and unlike charges attract.

Marks Awarded for**1** Law stated.

4(B)(b)**[2 marks]**

Define both the direction and magnitude of an electric field E at a point.

ANSWER

$E = F/q$; direction = direction of force on a positive charge

WORKING

Magnitude: the force per unit charge on a small test charge placed at the point, $E = F/q$ (unit $\text{N C}^{-1} = \text{V m}^{-1}$).

Direction: the direction of the force that a **positive** test charge would feel at that point.

Marks	Awarded for
1	Magnitude $E = F/q$.
1	Direction of force on a positive test charge.

4(B)(c)**[2 marks]**

An electron, released from rest, moves from left to right. What is the direction of E ? Does the electric potential energy of the particle increase or decrease as it moves?

ANSWER

E points right → left; potential energy decreases

WORKING

The electron is negative, so the force on it ($F = qE$ with $q < 0$) is **opposite** to E . It moves to the right, so E points **right to left**.

A charge released from rest moves from a region of higher electric PE to lower, just as a falling object loses gravitational PE. So its PE **decreases** (converted into kinetic energy).

Marks	Awarded for
1	E right to left.
1	PE decreases.

4(B)(d)(i)**[1 mark]**

A small water drop carrying 100 times the charge of an electron is prevented from falling by an electric field $E = 10^5 \text{ V m}^{-1}$. What is the direction of E ?

ANSWER

E points vertically downwards

WORKING

The electric force must point **upwards** to balance gravity. The drop is negative, so the force is opposite to E . E must therefore point **downwards**.

Marks	Awarded for
1	Downwards.

4(B)(d)(ii)**[4 marks]**

What is the mass of the drop?

ANSWER

$$m \approx 1.6 \times 10^{-13} \text{ kg}$$

WORKING

Charge on the drop: $q = 100 \times 1.602 \times 10^{-19} \text{ C} = 1.602 \times 10^{-17} \text{ C}$.

The drop is held still, so the electric force balances the weight:

$$q E = m g \Rightarrow m = q E / g$$

$$m = 1.602 \times 10^{-17} \times 10^5 \div 9.81 = 1.6 \times 10^{-13} \text{ kg}$$

Marks	Awarded for
1	$q = 100e$.
1	Force balance $qE = mg$.
1	Rearranged for m .
1	$\approx 1.6 \times 10^{-13} \text{ kg}$.

Question 4(B) Total: 10 Marks

Question 5 · Option A

Resistors in parallel, Ohm's law, ohmic and non-ohmic resistors — 10 marks

5(A)(a)

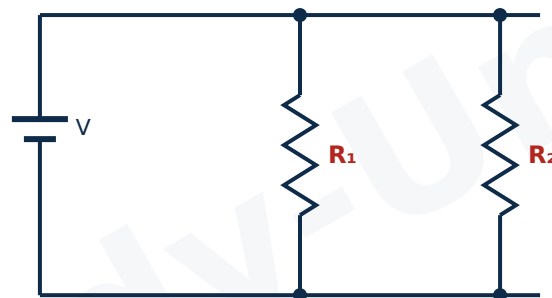
[2 marks]

Draw a simple electric circuit in which two resistors are connected in parallel, and write down an expression for their equivalent resistance.

ANSWER

$$1/R_{\text{eq}} = 1/R_1 + 1/R_2$$

WORKING



Both resistors are connected across the same two

Fig. 1 — Two resistors in parallel across a battery: each has the full battery voltage across it.

$$1 / R_{\text{eq}} = 1 / R_1 + 1 / R_2$$

Note on the lecture material: The PY1010 lecture notes don't cover combining resistors in parallel (or in series). The equivalent-resistance formula here comes from outside the lecture material.

Marks	Awarded for
1	Correct parallel circuit diagram.
1	Equivalent resistance expression.

5(A)(b)**[1 mark]**

Write down Ohm's Law.

ANSWER

$V = IR$

WORKING

$$V = IR$$

The current through a conductor is proportional to the potential difference across it, with a constant resistance R .

Marks Awarded for**1** $V = IR$.**5(A)(c)(i)****[1 mark]**

Resistors A ($100\ \Omega$) and B are in parallel across a 200 V battery, and 1 A flows through B. What is the voltage drop across B?

ANSWER

$V_B = 200\text{ V}$

WORKING

Each resistor is connected directly across the two battery terminals, so each has the full battery voltage across it.

Marks Awarded for**1** 200 V.

5(A)(c)(ii)**[1 mark]**

What is the resistance of B?

ANSWER

$$R_B = 200 \, \Omega$$

WORKING

$$R_B = V / I = 200 \, \text{V} \div 1 \, \text{A} = 200 \, \Omega$$

Marks**Awarded for****1**200 Ω .**5(A)(c)(iii)****[2 marks]**

What current flows through resistor A?

ANSWER

$$I_A = 2 \, \text{A}$$

WORKING

A also has 200 V across it:

$$I_A = V / R_A = 200 \, \text{V} \div 100 \, \Omega = 2 \, \text{A}$$

Marks**Awarded for****1**200 V across A, $I = V/R$.**1**

2 A.

5(A)(d)**[3 marks]**

When $V = 10 \text{ V}$, $I = 1 \text{ A}$; when $V = 20 \text{ V}$, $I = 1.5 \text{ A}$. Is R an ohmic or non-ohmic resistor? Why?

ANSWER

Non-ohmic: R changes from 10Ω to 13.3Ω (I is not proportional to V)

WORKING

Calculate $R = V/I$ at each setting:

$$R_1 = 10 \div 1 = 10 \Omega; R_2 = 20 \div 1.5 = 13.3 \Omega$$

For an ohmic resistor R is constant, so doubling V would double I (to 2 A). Here I rises only to 1.5 A and R changes, so the resistor does **not** obey Ohm's law.

Note on the lecture material: The lecture notes use $V = IR$ throughout but don't name "ohmic" and "non-ohmic" resistors. The answer applies the notes' $V = IR$ with R constant.

Marks	Awarded for
1	R at both settings.
1	States R is not constant / I not $\propto V$.
1	Non-ohmic.

Question 5(A) Total: 10 Marks

Question 5 · Option B

Lorentz force on a moving charge; electric and magnetic forces combined — 10 marks

Given: Positive charge $q = 1 \text{ mC}$ moves at $v = 300 \text{ m s}^{-1}$ along x ; B along y ; E along z .

5(B)(a)

[2 marks]

State and explain the formula for the magnitude of the Lorentz force on a charge moving through a magnetic field. State the SI units of all quantities.

ANSWER

$$\mathbf{F} = BQ\mathbf{v}; \mathbf{F} \text{ in N, } \mathbf{B} \text{ in T, } \mathbf{Q} \text{ in C, } \mathbf{v} \text{ in m s}^{-1}$$

WORKING

$$F = B Q v \text{ (for } v \text{ perpendicular to } B\text{)}$$

F: force (N). B: magnetic field (T). Q: charge (C). v: velocity (m s^{-1}). The force is perpendicular to both v and B .

Marks	Awarded for
1	$F = BQv$ ($v \perp B$).
1	All SI units.

5(B)(b)(i)**[3 marks]**

If $E = 10^3 \text{ V m}^{-1}$, what is the magnitude of the electric force on the particle? What is its direction?

ANSWER

$F_E = 1 \text{ N}$, along $+z$ (the direction of E)

WORKING

$$F_E = q E = 1 \times 10^{-3} \text{ C} \times 10^3 \text{ V m}^{-1} = 1 \text{ N}$$

The charge is positive, so the force points in the same direction as E : along $+z$.

Marks	Awarded for
1	$F = qE$.
1	1 N.
1	Along $+z$.

5(B)(b)(ii)**[3 marks]**

If $B = 2 \text{ T}$, what is the magnitude of the magnetic force on the particle? What is its direction?

ANSWER

$F_B = 0.6 \text{ N}$, along $+z$

WORKING

$$F_B = B q v = 2 \text{ T} \times 1 \times 10^{-3} \text{ C} \times 300 \text{ m s}^{-1} = 0.6 \text{ N}$$

The force is perpendicular to both v (along x) and B (along y), so it lies along z . For a positive charge with v along $+x$ and B along $+y$, the left-hand rule gives **$+z$** , the same direction as E .

Marks	Awarded for
1	$F = Bqv$.
1	0.6 N.
1	Along $+z$ (perpendicular to v and B).

5(B)(b)(iii)**[2 marks]**

Find the magnitude of the net force acting on the particle.

ANSWER

$F_{\text{net}} = 1.6 \text{ N (along +z)}$

WORKING

Both forces point along +z, so they add:

$$F_{\text{net}} = 1 \text{ N} + 0.6 \text{ N} = 1.6 \text{ N}$$

Marks	Awarded for
1	Both forces along the same axis, added.
1	1.6 N (own-figure with consistent directions accepted).

Question 5(B) Total: 10 Marks

Question 6 · Option A

Laws of reflection, total internal reflection, limit on a refractive index — 10 marks

6(A)(a)

[2 marks]

Write down the two laws of reflection.

ANSWER

$\theta_i = \theta_r$; incident ray, reflected ray and normal are coplanar

WORKING

(1) The angle of incidence equals the angle of reflection: $\theta_i = \theta_r$, both measured from the normal.

(2) The incident ray, the reflected ray and the normal all lie in the same plane.

Marks	Awarded for
1	$\theta_i = \theta_r$.
1	Same plane.

6(A)(b)**[3 marks]**

Using a diagram, describe what is meant by “total internal reflection”. Derive an expression for the critical angle θ_{crit} , describing any other quantities in it.

ANSWER

$\theta_{\text{crit}} = \sin^{-1}(n_2/n_1)$, with $n_1 > n_2$; above it, all light is reflected back

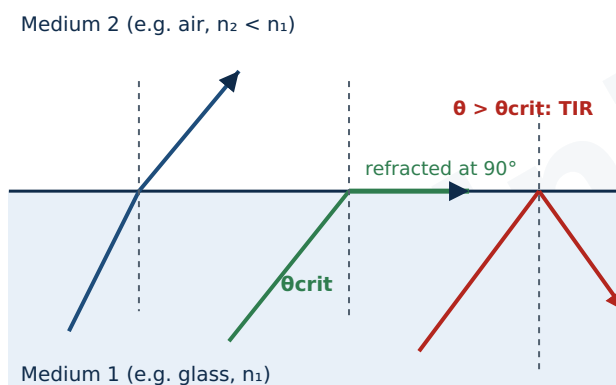
WORKING

Fig. 2 — Going from higher to lower n : below θ_{crit} the ray refracts out; at θ_{crit} it runs along the boundary; above θ_{crit} it is totally internally reflected.

When light travels from a medium of higher refractive index n_1 to one of lower index n_2 , it bends away from the normal. At the **critical angle** the refracted ray travels along the boundary ($\theta_2 = 90^\circ$). For larger angles of incidence the light is **totally reflected** back into medium 1.

Snell's law with $\theta_2 = 90^\circ$:

$$n_1 \sin \theta_{\text{crit}} = n_2 \sin 90^\circ = n_2 \Rightarrow \theta_{\text{crit}} = \sin^{-1}(n_2 / n_1)$$

n_1, n_2 are the refractive indices of the two media, with $n_1 > n_2$.

Marks	Awarded for
1	Diagram and description of TIR.
1	Derivation from Snell's law with $\theta_2 = 90^\circ$.
1	$\theta_{\text{crit}} = \sin^{-1}(n_2/n_1)$, terms described.

6(A)(c)**[5 marks]**

Glass ($n = 1.5$) rests on a second transparent medium. A ray in the glass meets the second medium at 50° incidence and undergoes total internal reflection. Calculate a limit on the refractive index of the second medium.

ANSWER

$$n_2 < 1.15$$

WORKING

TIR requires the angle of incidence to exceed the critical angle: $50^\circ > \theta_{\text{crit}}$. So $\sin 50^\circ > \sin \theta_{\text{crit}} = n_2/n_1$:

$$n_2 < n_1 \sin 50^\circ = 1.5 \times 0.766 = 1.15$$

(TIR is only possible at all if $n_2 < n_1$, which this satisfies.)

Marks	Awarded for
1	TIR condition: $\theta > \theta_{\text{crit}}$.
1	$\sin \theta_{\text{crit}} = n_2/n_1$.
1	Inequality set up correctly: $\sin 50^\circ > n_2/1.5$.
1	$n_2 < 1.5 \sin 50^\circ$.
1	$n_2 < 1.15$ (an upper limit).

Question 6(A) Total: 10 Marks

Question 6 · Option B

Huygens' principle, Young's double slit, interference — 10 marks

Given: Red laser $\lambda = 650 \text{ nm}$; slit separation $3.73 \mu\text{m}$; point P at 5° to the beam.

6(B)(a)

[2 marks]

Describe how Huygens' Principle can be used to describe the propagation of a wave as it meets an obstacle.

ANSWER

Each point on a wavefront is a source of wavelets; their envelope is the new wavefront, so waves spread around obstacles (diffraction)

WORKING

Every point on a wavefront acts as a source of new spherical wavelets. The new wavefront is the surface that touches (envelops) all these wavelets.

At an obstacle or opening, only the unblocked points on the wavefront produce wavelets. These spread out into the region behind the obstacle, so the wave bends around edges (diffraction).

Note on the lecture material: Huygens' principle is **not covered in the PY1010 lecture notes**. This answer is included for completeness.

Marks	Awarded for
1	Each point on a wavefront is a source of wavelets.
1	Applied to an obstacle: the wave spreads/bends around it.

6(B)(b)**[3 marks]**

Describe, with the aid of a diagram, Young's double slit experiment.

ANSWER

Two in-phase slit sources superpose to give bright (constructive) and dark (destructive) fringes on a screen

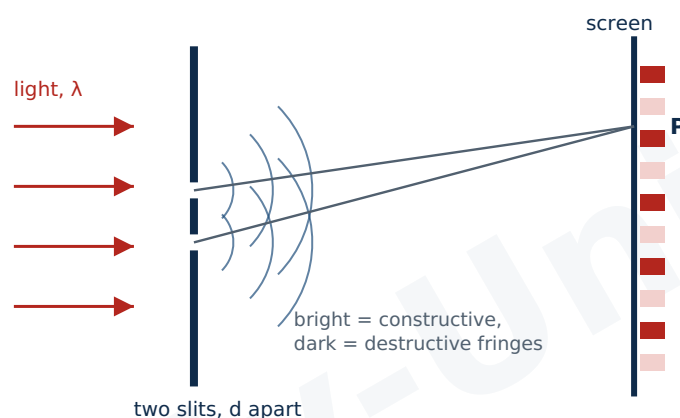
WORKING

Fig. 3 — Light from one source passes through two close slits; the two waves overlap and give bright and dark fringes on a screen.

Light of a single wavelength falls on two narrow, closely spaced slits. The slits act as two sources **in phase** with each other.

The waves from the two slits overlap and **superpose** on a screen. Where they arrive in phase they interfere constructively (bright fringes); where they arrive exactly out of phase they interfere destructively (dark fringes). The result is a pattern of alternating bright and dark bands, which shows the wave nature of light.

Note on the lecture material: Young's double slit experiment is **not covered in the lecture notes**. The superposition and interference ideas used are, but the experiment itself comes from outside the lecture material.

Marks	Awarded for
1	Labelled diagram: source, two slits, screen.
1	Two coherent (in-phase) sources superpose.
1	Bright/dark fringes explained by constructive/destructive interference.

6(B)(b)(i)**[1 mark]**

The phase difference between the waves from each slit arriving at P is 2λ .
Constructive or destructive interference? Explain.

ANSWER

Constructive: 2λ is a whole number of wavelengths, so the waves are in phase

WORKING

A difference of 2λ is a whole number of wavelengths, so crest meets crest and the waves arrive **in phase**. Their amplitudes add.

Marks	Awarded for
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1	Constructive, with reason.
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6(B)(b)(ii)**[1 mark]**

If the phase difference is $\frac{3}{2}\lambda$, what type of interference is observed? Explain.

ANSWER

Destructive: 1.5λ puts the waves exactly out of phase

WORKING

$\frac{3}{2}\lambda = 1.5\lambda$ is an odd number of half-wavelengths, so crest meets trough and the waves are **exactly out of phase**. They cancel.

Marks	Awarded for
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1	Destructive, with reason.
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6(B)(c)(i)**[2 marks]**

(Printed as a second “(b)” on the paper.) The line from the slits to P makes 5° with the original beam direction. Calculate the path difference between the waves from the slits at P, in terms of λ .

ANSWER

Path difference = 0.5λ ($\lambda/2$)

WORKING

The extra distance travelled by the wave from the further slit is $d \sin \theta$:

$$\Delta = d \sin \theta = 3.73 \times 10^{-6} \text{ m} \times \sin 5^\circ = 3.25 \times 10^{-7} \text{ m}$$

$$\Delta / \lambda = 3.25 \times 10^{-7} \div 6.50 \times 10^{-7} = 0.50$$

Note on the lecture material: The path-difference formula $d \sin \theta$ is not in the lecture notes (it belongs with Young's experiment above).

Marks	Awarded for
1	$\Delta = d \sin \theta$.
1	0.5λ .

6(B)(c)(ii)**[1 mark]**

What type of interference do you expect in this case?

ANSWER**Destructive interference (dark fringe)****WORKING**

A path difference of $\lambda/2$ means the waves arrive exactly out of phase, so they cancel: a dark fringe at P.

Marks	Awarded for
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1	Destructive.
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Question 6(B) Total: 10 Marks